

MA140: Sample Test 1 10:00, Tuesday 99 October, 2099

Your ID number							

Your name:

.....

Instructions

- Duration: **40 minutes**
- You may use a calculator.
- You may **not** use notes or any electronic device (other than a non-programmable calculator).
- Each question has **one** correct answer.
- Use a dark coloured pen or pencil and completely fill answer boxes corresponding to your choice. Shade each box fully:

Good



Bad

*Some formulae*

$$f'(a) := \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}. \quad \frac{d}{dx} x^n = nx^{n-1}.$$

First: encode your ID number in the boxes on the left. The 1st digit of your ID number goes in the 1st column, the 2nd digit in the 2nd column, etc.

Question 1 Let $f(x) = \begin{cases} x+2 & x \leq -2 \\ -x & x > 2. \end{cases}$ Why type of *discontinuity* does f have at $x = 2$?

☐ Jump discontinuity☐ Infinite discontinuity☐ None (f is continuous)☒ Removable discontinuity

Question 2 Let $f(x) = 2x^3 - 3x^2$. Which *one* of the following statements is true?

☐ $f'(0) = 1$ ☐ $f'(-2) = 12$ ☒ $f'(1) = 0$ ☐ $f'(1) = -1$

Question 3 Suppose that $g(x) = 2x^4 + x^2$, and $f(x)$ is such that $-g(x) \leq f(x) \leq g(x)$ for all x . Which *one* of the following statements is true?

☐ One can use the Squeeze Theorem to determine $\lim_{x \rightarrow 1} f(x)$ ☐ One cannot use the Squeeze Theorem to determine $\lim_{x \rightarrow 0} f(x)$ ☒ One can use the Squeeze Theorem to determine $\lim_{x \rightarrow 0} f(x)$

Question 4 What is the largest possible subset of $\mathbb{R}$ that could be the **domain** of the function:

$$f(x) = \frac{2x}{x^2 + 2x - 8}?$$

☐ $(-\infty, -4) \cup (-4, 0) \cup (0, 2) \cup (2, \infty)$

☒ $(-\infty, -4) \cup (-4, 2) \cup (2, \infty)$

☐ $x = 2$ and $x = -4$

☐ all of $\mathbb{R}$

Question 5 Let $f(x) = \frac{x^2 - 2x - 15}{3x^3 - 6x^2 - 45x}$. Which *one* of the following statements is true?

☐ $\lim_{x \rightarrow 0^+} f(x) = -\infty$

☒ $\lim_{x \rightarrow 0^-} f(x) = -\infty$

☐ $\lim_{x \rightarrow 0} f(x)$ exists

☐ $\lim_{x \rightarrow 0^+} f(x) = 0$

Question 6 What does $\lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{x^2 + x - 12}$ evaluate as?

☒ $\frac{5}{7}$

☐ $-\infty$

☐ $-\frac{7}{5}$

☐ 0

☐ ∞

Question 7 Let $f(x) = \frac{1}{x^2 - 1}$. Which *one* of the following statements is correct?

☐ f has one vertical asymptote, which is at $x = 1$

☐ f does not have any vertical asymptotes

☒ f has two vertical asymptotes, which are at $x = 1$ and $x = -1$

☐ f has one vertical asymptote, which is at $x = -1$

Question 8 Let $f(x) = -4x^3 + 2x^2 - 4$. Which *one* of the following is the tangent to f at $x = -1$?

☐ $y = -8x - 6$

☐ $y = -4$

☐ $y = -8x + 2$

☒ $y = -16x - 14$

Question 9 Which of the following corresponds to $f(x) = \frac{8x - 12}{x^2 - 2x - 3}$ expressed as *partial fractions*?

☐ $\frac{3}{x+3} - \frac{5}{x-1}$

☐ $\frac{3}{x-3} - \frac{5}{x+1}$

☒ $\frac{3}{x-3} + \frac{5}{x+1}$

☐ $\frac{3}{x+3} + \frac{5}{x-1}$